Integrons: agents of bacterial evolution

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Integrons are assembly platforms — DNA elements that acquire open reading frames embedded in exogenous gene cassettes and convert them to functional genes by ensuring their correct expression. They were first identified by virtue of their important role in the spread of antibiotic-resistance genes. More recently, our understanding of their importance in bacterial genome evolution has broadened with the discovery of larger integron structures, termed superintegrons. These DNA elements contain hundreds of accessory genes and constitute a significant fraction of the genomes of many bacterial species. Here, the basic biology of integrons and superintegrons, their evolutionary history and the evidence for the existence of a novel recombination pathway is reviewed.

Transposon
A mobile DNA element that can relocate within the genome of its host.

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doi:10.1038/nrmicro1462

The emergence and spread of antibiotic resistance among bacterial pathogens is the most striking example of evolution that has been observed in bacteria over the past six decades. Indeed, antimicrobial resistance has been an impediment to effective infectious disease therapy for as long as antibiotics have been used. Single-drug-resistance phenotypes were not entirely unforeseen, as was demonstrated in early laboratory studies. By contrast, multidrug resistance was not anticipated, because the co-appearance of multiple mutations conferring resistance was considered to be beyond the evolutionary potential of a given bacterial population. However, only six years after the introduction (and massive production) of streptomycin, tetracycline and chloramphenicol, isolates of *Shigella dysenteriae* that were simultaneously resistant to each of these antibiotics, and to sulphonamides, were identified[1]. At this time, it also became clear that the emergence of multidrug-resistant strains could not be attributed to mutation alone. In the 1970s, multidrug resistance was determined in many cases to be associated with transmissible plasmids, and the importance of integrons in the acquisition of resistance genes was only recognized much later, in the late 1980s (REF. 2.) It is clear, however, that integrons contributed to the initial outbreaks of multidrug resistance in the 1950s, as indicated by the involvement of Tn*21*, an integron-containing transposon, in the resistance phenotype propagated by plasmid NR1 (R100)[3]. Given the timescale of the development of multidrug-resistant *Shigella* strains, it is clear that bacteria were prepared for such a challenge and had already evolved the appropriate genetic tools, including integrons.

What is an integron?

Integrons are assembly platforms that incorporate exogenous open reading frames (ORFs) by site-specific recombination and convert them to functional genes by ensuring their correct expression. All integrons characterized to date are composed of three key elements necessary for the capture of exogenous genes: a gene (*intI*) encoding an integrase belonging to the tyrosine-recombinase family; a primary recombination site (*attI*); and an outward-orientated promoter (P_c) that directs transcription of the captured genes[4]. Integron-encoded integrases can recombine discrete units of circularized DNA[5] known as gene cassettes in a RecA-independent manner. Integration occurs downstream of the resident P_c promoter at the *attI* site, allowing expression of the genes in the cassette (FIG. 1). All integron-inserted gene cassettes identified share specific structural characteristics and generally contain a single gene and an imperfect inverted repeat at the 3′ end of the gene called an *attC* site (or 59-base element). The *attC* sites are a diverse family of nucleotide sequences that function as recognition sites for the site-specific integrase. They vary in length from 57 bp to 141 bp[6], and their nucleotide sequence similarities are primarily restricted to the boundaries, which contain conserved sequences known as the R″ sequence (RYYYAAC, where R is a purine and Y is a pyrimidine) and the R′ sequence[7] (GTTRRRY, where the point of recombination is between the G and the T bases) (FIG. 1).

When integrons were first described, the definition of an integron suggested that the integron element itself was a mobile DNA element[2]. This assumption stemmed from the fact that the first integrons to be characterized were located in transposons. In these examples, the

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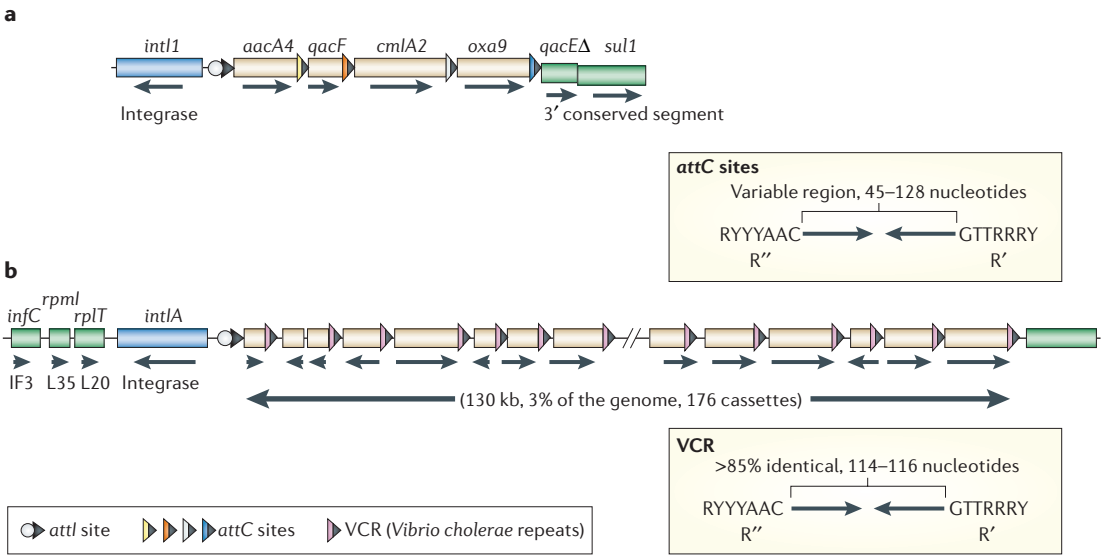

Figure 1 | **Mobile integrons and superintegrons.** Structural comparison of a 'classical' mobile integron and the superintegron from *Vibrio cholerae* strain N16961. **a** | A schematic representation of the class 1 integron In40. The various resistance-gene cassettes carry different *attC* sites. The following antibiotic-resistance cassettes confer resistance to the following compounds: *aacA4*, aminoglycosides; *cmlA2*, chloramphenicol; *oxa9*, β-lactams; *qacF and qacE*, quarternary ammonium compounds. The *sul* gene, which confers resistance to sulphonamides, is not a gene cassette. **b** | A schematic representation of the chromosomal superintegron in *V. cholerae*; the open reading frames are separated by highly homologous sequences, the *V. cholerae* repeats (VCRs). *infC*, encodes translation initiation factor IF3; *rpmI* and *rplT* encode ribosomal proteins L35 and L20, respectively.

transposition of the integrons did not depend on the activity of the integron-encoded integrase, which only mobilizes the gene cassettes within integrons. However, with the discovery of other types of integron, either carried by transposons or present as immobile components of bacterial genomes, the definition of an integron has evolved towards the definition outlined at the start of this section. All integrons can be divided into two distinct subsets: the mobile integrons, which are linked to mobile DNA elements and are primarily involved in the spread of antibiotic-resistance genes; and the superintegrons.

Mobile integrons

At present, five classes of mobile integrons are known to have a role in the dissemination of antibiotic-resistance genes. These classes have been historically defined based on the sequence of the encoded integrases, which show 40–58% identity. All five classes are physically linked to mobile DNA elements, such as insertion sequences (ISs), transposons and conjugative plasmids, all of which can serve as vehicles for the intraspecies and interspecies transmission of genetic material. Three classes of mobile integrons are 'historical' classes that are involved in the multiple-antibiotic-resistance phenotype[8]. Class 1 integrons are associated with functional and non-functional transposons derived from Tn*402* (REFS 9,10) that can be embedded in larger transposons, such as Tn*21*. Class 2 integrons are exclusively associated with Tn*7* derivatives[9,11], and class 3 integrons are thought to be located in a transposon[12] inserted in as-yet-uncharacterized plasmids[13–15]. The other two classes of mobile integrons, class 4 and class 5, have been identified through their involvement in the development of trimethoprim resistance in *Vibrio* species; one (class 4) is a component of a subset of SXT elements found in *Vibrio cholerae*[16], and the other (class 5) is located in a compound transposon carried on a plasmid in *Vibrio salmonicida* (H. Sørum *et al.*, unpublished observations).

Class 1 integrons are found extensively in clinical isolates, and most of the known antibiotic-resistance-gene cassettes belong to this class. To date, and considering only those cassettes that differ in nucleotide sequence by more than 5%, over 80 different gene cassettes from class 1 integrons have been described. Between them, these elements confer resistance to all known β-lactams, all aminoglycosides, chloramphenicol, trimethoprim, streptothricin, rifampin, erythromycin, fosfomycin, lincomycin and antiseptics of the quaternary-ammonium-compound family (reviewed in REFS 17,18). By contrast, only six different resistance cassettes have been found that are associated with class 2 integrons[19,20]. This reduction in diversity is probably owing to the fact that the gene encoding the integrase in class 2 integrons contains a nonsense mutation in codon 179 (ochre 179), thereby yielding a truncated, non-functional protein[21]. Mutation of the ochre 179 codon into a glutamic-acid-encoding codon is sufficient to produce an integrase with full recombinase activity[21]. However, it is not known whether the differences in cassette content in the different Tn*7* derivatives are due to occasional natural suppression of the ochre 179 codon, leading to an active integrase, or due to the *trans*-acting recombination activity of another integrase, such as the class 1 integrase (IntI1),

Insertion sequence
(IS) A small (< 2.5 kb), generally phenotypically cryptic segment of DNA that has a simple organization and is capable of insertion at multiple sites in a target DNA molecule. Examples include IS*1*, IS*608* and IS*911*.

Conjugative plasmid
A plasmid that can move from one cell to another during the process of conjugation.

SXT element
Vibrio cholerae-derived integrating and conjugative element (also referred to as a conjugative transposon or constin).

Compound transposon
A segment of DNA flanked by two similar insertion sequences, in direct or inverted orientations. Examples include Tn*5* and Tn*10*.

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which has been shown to recognize and recombine the *attI* site of class 2 integrons (*attI2)* (REF. 21). It has also been shown that IntI1 can recombine diverse *attC* sites[22]. The IntI integrases belong to the catalytic family of tyrosine recombinases that are involved in the horizontal movement of numerous phage (including the λ phage integrase, λInt) and in fundamental cellular processes, such as the resolution of chromosome dimers during cell division. Unlike the other members of the tyrosine-recombinase family, however, IntI integrases can recombine nucleotide sequences of low similarity.

Recently, a new term — complex *sul1*-type integrons — has been introduced to describe the large genetic elements that associate different class 1 integrons with *orf513*. This is a gene encoding a transposase belonging to the IS*91* family, which is involved in the recruitment of novel resistance genes (see REF. 23 for an example) and has been found in *Salmonella enterica* serovar Newport[24]. The name 'complex integron' is an unfortunate choice as it suggests that these elements are a new kind of integron, with a different structure and machinery, when in fact this is not the case; the novelty in the gene-recruitment aptitude of these structures is exclusively linked to the activity of *orf513*.

The efficiency of the partnership between integrons and mobile DNA elements is confirmed by the differences in codon usage among gene cassettes in the same mobile integron, indicating that the acquired genes have diverse origins. By using integrons, bacteria can stockpile different exogenous genes to establish an appreciable antimicrobial-resistance 'armamentarium', and mobile integrons harbouring up to eight different resistance cassettes have been characterized[25]. Mobile integrons carrying resistance cassettes have been identified in diverse Gram-negative species, but they are not restricted to these microorganisms, as class 1 integrons have also been found in Gram-positive bacteria, including corynebacteria[26,27], aerococci, brevibacteria and staphylococci[28].

Chromosomal superintegrons

In the late 1990s, studies examining the relationship between the captured gene cassettes in mobile integrons and a cluster of repeated DNA sequences with a similar organization identified on chromosome 2 of the *V. cholerae* genome (called VCRs for *V. cholerae* repeats[29]) led to the discovery of a distinct type of integron[30]. This integron encodes a specific integrase, VchIntIA, which is related to the integrases encoded by mobile integrons but has two characteristics that distinguish it from known mobile integrons. First, there are a large number of gene cassettes that are associated with the integron, and there is a high degree of identity (>80%) observed between the *attC* sites of these cassettes (the VCRs; FIG. 1). Second, the structure does not seem to be mobile — the integron is located on the chromosome and is not associated with mobile DNA elements[31]. These key features define the superintegron subset.

Superintegrons are now known to be integral components of many γ-proteobacterial genomes, and they have been identified in the Vibrionaceae and their close relatives the xanthomonads, and in a branch of the pseudomonads (TABLE 1). They share the same general characteristics: they are large and carry >20 gene cassettes, and there is extensive homology between the *attC* sites of their endogenous cassettes. The superintegrons in the genomes of four *Vibrio* species have been sequenced, and all contain a large number of gene cassettes, ranging from 72 associated with the superintegron of *Vibrio parahaemolyticus*[32] to more than 200 in *Vibrio vulnificus*[33]. In some cases, the superintegron gene-cassette array is split into several segments located on different chromosomes — two in the case of *Vibrio fischeri* strain ES114 (REF. 34) and four in the case of *Photobacterium profundum* strain SS9 (accession numbers NC_006370 and NC_006371 (REF. 35)), in addition to a few cases where isolated gene cassettes are present on both chromosomes (D.M., unpublished observations). Incidentally, the *P. profundum intIA* gene has been inactivated by a deletion that has removed two-thirds of the C-terminal region of the protein (TABLE 1). It is also likely that splitting of the superintegron gene-cassette array occurred in *Pseudomonas alcaligenes*[36].

The superintegron of *V. cholerae* serotype O1 biotype El Tor strain N16961 contains at least 215 ORFs (mostly unidentified, VCA0291–VCA0506) in an array of 179 gene cassettes that begins at the gene encoding VchIntIA and occupies ~3% of the genome[37]. Sequence analysis showed that the *attC* sites carried by 149 of the 179 cassettes differed by less than 10% over their entire length of 122 to 124 nucleotides[37]. Similar analyses performed on the superintegron cassette arrays from other bacterial species (*V. fischeri*, *Vibrio metschnikovii*, *P. alcaligenes*, *Pseudomonas stutzeri* and *Xanthomonas campestris*) have shown that this characteristic is conserved[36–39]. The high level of species-specific sequence identity shared by the majority of the *attC* sites in each superintegron suggests that gene cassettes are assembled through the physical association of a superintegron *attC* site with an incoming DNA fragment. The mechanism(s) responsible for this process is unknown.

Several integrons have been characterized in the genomes of different bacterial species that do not share all of the characteristics of the typical superintegron. For example, the superintegrons in the genomes of two *Shewanella* species contain only a small number of cassettes (three in the case of *Shewanella oneidensis*[40]) with structurally different *attC* sites, a situation similar to the gene-cassette arrays associated with mobile integrons. The integron in the genome of *Nitrosomonas europaea* does not contain any gene cassettes. However, experiments indicate that the integrases and cognate *attI* sites from each of these species are functional[40,41]. Furthermore, ORFs that are homologous to the genes encoding integron-associated integrases have also been identified in the genomes of δ- and β-proteobacteria, and in bacterial species belonging to other taxa (for example, the marine planctomycete *Rhodopirellula baltica* SH 1; also known as *Pirellula* sp. strain 1); however, these hypothetical integrases have not been characterized (TABLE 1). An element containing all of the typical superintegron features but with an unusual variation in its organization has also been found in the genome of the

Transposase
The enzyme that promotes cutting of the DNA at the ends of a transposable element and joining to the DNA molecule into which the element is to be inserted.

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Table 1 | **Bacterial species harbouring chromosomal integrons and superintegrons**

Bacterial group	Bacterial strain	IntIA accession number	Characteristics
γ-Proteobacteria — Vibrionaceae and close relatives	*Vibrio cholerae*	AAC38424	Aetiological agent of cholera in humans
	V. mimicus	AAD55407	Certain serogroups are enterotoxic human pathogens
	V. metschnikovii	AAK02074	
	V. parahaemolyticus	AAK02076	Certain strains can cause seafood-borne gastroenteritis in humans
	V. splendidus		Strains are mutualistic, opportunistic and pathogenic for marine animals
	V. harveyi		Pathogen of black tiger prawns
	V. natriegens	AAO38263	
	V. hollisae		Certain strains can cause gastroenteritis in humans
	V. salmonicida		Marine fish pathogen
	V. fischeri	Strain CIP 103206, AAK02079; strain ES114, AAW87733	Non-pathogenic, luminescent bacterium
	V. vulnificus	Strain CIP 75.4, AAN33109; strain CMCP6, AAO10775; strain YJ016, BAC94705	Human and animal pathogen causing septicaemia
	Listonella anguillarum	AAM95157	Marine fish pathogen
	L. pelagia	AAK02082	Marine bacterium that produces tetrodotoxin
	Alteromonas macleodii		Marine bacterium
	Photobacterium profundum SS9	CR378678 (translation from 312617 to 312905)	Luminescent bacterium
	Moritella marina		Psychrophilic marine bacterium
— *Shewanella*	*Shewanella oneidensis*	AAN55084	Diverse genus of bacteria well known for their ability to use dissimilar compounds as electron acceptors; prominent in fish spoilage and hull oxidation in the marine industry
	S. putrefaciens	AAK01408	
	S. amazonensis SB2B	EAN38747	
	Shewanella sp. MR-7	EAP24271	
— Xanthomonadaceae	*Xanthomonas campestris* pv. *campestris*	AAK07444	Gram-negative phytopathogenic species responsible for disease in virtually all major taxa of plant life
	X. campestris pv. *badrii*	AAK07443	
	Xanthomonas sp. 102397	AAK07447	
	Xanthomonas sp. 102336		
	Xanthomonas sp. 102338		
	Xanthomonas sp. 105155		
	X. oryzae		
— Pseudomonadaceae	*Pseudomonas pseudoalcaligenes*		Genus includes opportunistic pathogens
	P. alcaligenes	AAK73287	
	P. mendocina		
	P. stutzeri	Strain BAM, AAN16071; strain Q, AAN16061	
	Pseudomonas sp. NEB 376		
— Alteromonadaceae	*Saccharophagus degradans*	ZP_00318025	Marine bacterium capable of degrading insoluble complex polysaccharides
— Ectothiorhodospiraceae	*Nitrococcus mobilis* Nb-231	EAR20185	
— Various	*Reinekea* sp. MED297	EAR08903	
β-Proteobacteria	*Nitrosomonas europaea*	CAD84361; CAD86100	Soil bacterium
	Thiobacillus denitrificans	YP_314973	Capable of autotrophic denitrification
	Azoarcus sp. EbN1	YP_157034	Anaerobic aromatic-degrading denitrifying bacterium
	Methylobacillus flagellatus KT	YP_544521	Obligate methylotroph
	Dechloromonas aromatica RCB	YP_286626	Capable of anaerobic benzene oxidation
	Rubrivivax gelatinosus PM1	ZP_00244730	Facultative phototrophic non-sulphur bacterium
δ-Proteobacteria	*Geobacter metallireducens* GS-15	AAB33221	Fe(III)-reducing bacteria
	G. sulfurreducens	NP_953513	
Planctomycetes	*Rhodopirellula baltica* SH 1	NP_865348	Marine strain
Spirochaetales	*Treponema denticola*	NP_972448	Causes periodontal disease

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spirochaete *Treponema denticola* ATCC 35405 (REF. 42). This integron has its *intIA* gene in the inverse orientation compared with the canonical structure — the *attI* site is located adjacent to the 3′ end of the *intIA* gene, followed by an array of 45 gene cassettes containing highly related *attC* sites. In all other cases, the *attI* site is located adjacent to the 5′ end of the *intIA* gene (FIG. 1).

The ubiquitous nature of the integrons in the genomes of environmental bacterial species has been confirmed by the detection of genes encoding integron-associated integrases and gene cassettes from environmental DNA samples taken from markedly different sources. Using PCR primers directed against conserved regions of the integron integrase genes and *attC* sites, the groups of Stokes and Schmidt were able to amplify 19 new integron-encoded integrases and hundreds of gene cassettes[43–46]. In most cases, the protocol used did not allow determination of the source of the integrons, whether endogenous superintegrons from soil bacteria or integrons located on mobile genetic elements.

Integrons are ancient evolutionary structures

Comparative analysis of all the integrases associated with integrons, including both characterized and hypothetical examples, shows that they group together and form a specific clade within the tyrosine-recombinase family[38,46,47] (FIG. 2). Additionally, all integron-associated integrases contain a conserved sequence of 16 amino acids[43,48] located between domains II and III (REF. 49). Integrons are undoubtedly ancient entities, as indicated by the species-specific clustering of the respective superintegron integrase genes in a pattern that adheres to the line of descent among the bacterial species in which they are located (as in the Vibrionaceae[37] and Xanthomonadaceae[39] radiations). Therefore, the establishment of superintegrons probably pre-dates speciation in the respective genera, indicating that integrons are ancient structures that have been impacting on the evolution of bacterial genomes for hundreds of millions of years. Even if superintegron platforms are not associated with an identifiable mobile DNA element, it remains possible that horizontal transfer of part or all of the structure occurred over the course of a long period of evolution. Indeed, the process of horizontal gene transfer could be the cause of the discrepancies observed between the superintegron integrase gene tree and the 16S rRNA gene tree generated for *V. fischeri*[38,46]. Horizontal gene transfer could also be responsible for the presence of the many *intI*-like genes in the genome sequences of different δ- and β-proteobacteria. Further analysis should reveal whether these *intI*-like genes are part of mobile elements.

It is also interesting to note that the phylogenetic congruency in the tree topologies for the *intI* and 16S rRNA sequences is consistent with the respective environmental niches of the host microorganisms (FIG. 2). Indeed, the integron-associated integrases from marine bacteria all group together, and those from freshwater or soil bacteria (whether β-, δ- or γ-proteobacteria) are closely related to one another. This is probably a reflection of the extent of DNA exchange among bacterial species occupying similar environments.

Biological functions associated with integrons

Other than the antibiotic-resistance genes associated with mobile integrons, a precise inventory of the functions encoded by integron and superintegron gene cassettes remains to be compiled. The majority of the superintegron gene cassettes examined so far seem to be unique to the host species and have no known function. A known antibiotic-resistance gene cassette associated with a mobile integron from a clinical isolate has yet to be identified in a superintegron. However, several superintegron cassettes encoding genes that have significant homology to aminoglycoside-, phosphinotricin-, fosfomycin- and streptothricin-resistance genes have been identified and might have the potential to express a resistance phenotype, if exposed to the appropriate drug-selection pressures[50]. Recently, two such genes from superintegron cassettes located in the genomes of environmental isolates were expressed as recombinant proteins and the enzyme activity assayed[51]. One of these proteins, which showed sequence similarity to an aminoglycoside phosphotransferase (APH), displayed an ATPase activity that was consistent with the presence of Mg^{2+}-binding residues (a characteristic of APH proteins)[52]. However, its substrate is still unknown. The second protein showed sequence similarity to the RNA methyltransferase superfamily, and the recombinant protein did exhibit methyltransferase activity with RNA.

Preliminary studies indicate that superintegron gene cassettes also encode proteins involved in adaptive functions other than antibiotic resistance. In *V. cholerae*, genes encoding several virulence factors, including the heat-stable toxin gene (*sto*), the mannose-fucose-resistant haemagglutinin gene (*mrhA*) and a lipoprotein gene, are located in superintegron gene cassettes[53–55]. In *V. vulnificus*, a gene identified through a transposon-mutagenesis screen as being essential for the expression of the main *V. vulnificus* virulence determinant, the capsular polysaccharide, is also present in a superintegron gene cassette[56]. The metabolic functions of three proteins encoded by superintegron gene cassettes — a *V. cholerae* sulphate-binding protein, a psychrophilic lipase in *Moritella marina* and a restriction enzyme (*Xba*I) and its cognate methylase from *X. campestris* pathovar *badrii* — have all been determined experimentally[38]. Additional proteins encoded by superintegron gene cassettes include proteins that have homology to DNA-modification enzymes (including nudix hydrolases and restriction endonucleases) and enzymes involved in primary metabolism[36,37,50,57].

Of particular interest is the discovery that superintegrons carry several gene cassettes encoding members of different toxin–antitoxin families found in low-copy-number plasmids[37,58]. These cassettes seem to be among the rare cassettes that carry their own promoter and are therefore expressed outside the context of *attI* and the P_c promoter control. They might have an important role in stabilizing the large array of cassettes found in the superintegrons in *Vibrio* spp., as their loss is deleterious for the host genome[37].

Transposon-mutagenesis screen
The use of transposons to generate (knock-out) mutations.

Nudix hydrolases
Nudix hydrolases (nucleoside diphosphate linked to some other moiety X) are enzymes that hydrolyse diverse nucleoside diphosphate or triphosphate derivatives.

Toxin–antitoxin families
Paired loci found in the chromosomes of almost all free-living bacteria, and many plasmids and phage genomes. They encode a toxin and its antidote, which have been shown to contribute to plasmid stability by a mechanism called post-segregational killing and are also proposed to function in bacterial programmed cell death or stress physiology.

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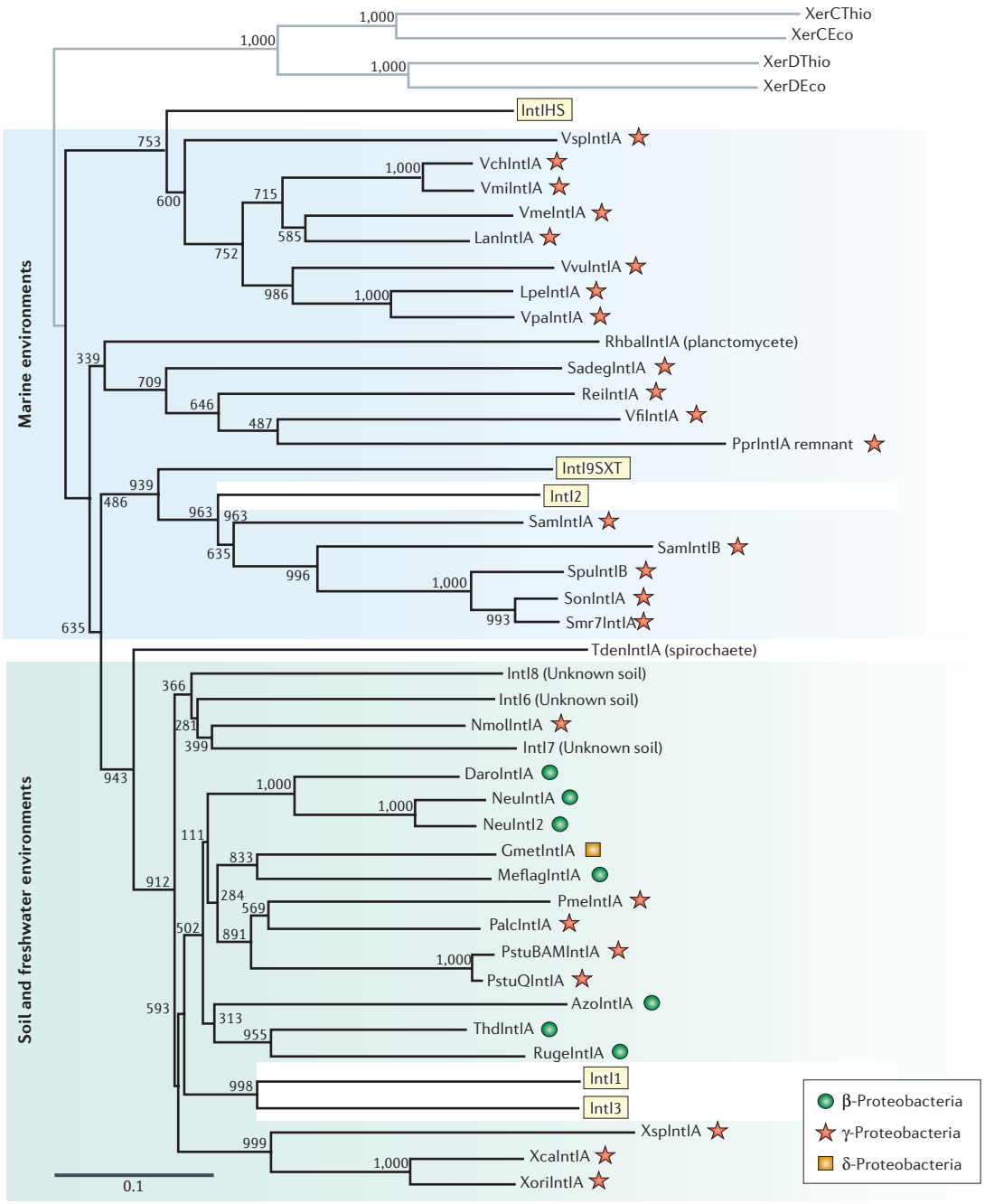

Figure 2 | **Phylogenetic relationship of the integron *intI* genes in the proteobacteria.** The dendrogram is based on known *intI* gene sequences. The tree was rooted using XerC and XerD from *Escherichia coli* (XerCEco and XerDEco, respectively) and *Thiobacillus denitrificans* ATCC 25259 (XerCThio and XerDThio, respectively). The integrases from the five classes of mobile integrons are boxed. The accession numbers for IntIHS and IntI9SXT are AJ277063 and AY035340, respectively. The sources of IntI6, IntI7 and IntI8 are unknown. The tree displayed is the best-distance neighbour-joining tree obtained using MEGA3. Bootstrap support values represent the consensus of distance neighbour-joining trees obtained from 1,000 pseudo-replicates of the data set. Branch lengths were drawn proportional to the amount of evolution based on genetic distance; the scale bar indicates the number of substitutions per site. Accession numbers (where available) can be found in TABLE 1. Abbreviations for the organisms in which the integron integrases can be found are as follows: Azo, *Azoarcus* sp. EbN1; Daro, *Dechloromonas aromatica*; Eco, *E. coli*; Gmet, *Geobacter metallireducens*; Lan, *Listonella anguillarum*; Lpe, *Listonella pelagia*; Meflag, *Methylobacillus flagellatus*; Neu, *Nitrosomonas europaea*; Nmo, *Nitrococcus mobilis*; Palc, *Pseudomonas alcaligenes*; Pme, *Pseudomonas mendocina*; Ppr, *Photobacterium profundum*; PstuBAM, *P. stutzeri* BAM; PstuQ, *Pseudomonas stutzeri* Q; Rei, *Reinekea* sp.; Rhbal, *Rhodopirellula baltica*; Ruge, *Rubrivivax gelatinosus*; Sadeg, *Saccharophagus degradans*; Sam, *Shewanella amazonensis*; Smr7, *Shewanella* sp. MR-7; Son, *Shewanella oneidensis*; Spu, *Shewanella putrefaciens*; Tde, *Treponema denticola*; Thd, *Thiobacillus denitrificans*; Vch, *Vibrio cholerae*; Vfi, *Vibrio fischeri*; Vme, *Vibrio metschnikovii*; Vmi, *Vibrio mimicus*; Vpa, *Vibrio parahaemolyticus*; Vsp, *Vibrio splendidus*; Vvu, *Vibrio vulnificus*; Xca, *Xanthomonas campestris*; Xori, *Xanthomonas oryzae*; Xsp, *Xanthomonas* spp.

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The determination of the diverse metabolic activities associated with superintegron gene cassettes (other than antibiotic resistance and virulence) indicates that integrons operate as a general gene-capture system in bacterial adaptation. Analysing the biological diversity of the captured genes has the potential to lead to the discovery of novel proteins of scientific and biotechnological value.

The level and regulation of expression of superintegron gene cassettes is largely unknown. Examples in which gene expression has been demonstrated include the heat-stable enterotoxin, *ccdAB*, and CARB-7 and CARB-9 cassettes (see later). In other instances, no expression could be detected. For example, it has been demonstrated that there is no expression of the *catB9* superintegron cassette in *V. cholerae* strain N16961 (REF. 59). However, a recent comparative transcriptome analysis of this strain, and analysis of three derivatives with mutations in either *hapR* (which encodes a regulator of the virulence factor TcpP) or *rpoS* or *rpoN* (which both encode alternative σ-factors), indicates that the transcription of the majority of the genes present in the superintegron gene cassettes of *V. cholerae* N16961 is induced at high cell densities and in response to stress or in stationary phase[60]. From an evolutionary point of view, this observation supports a link between the gene content of the superintegron cassettes and the adaptive capacity of the bacterial host, especially in the context of a survival-enhancing stress response.

Variation in superintegron gene cassettes

A comparison of the gene cassettes in different *Vibrio* species showed that the majority of the cassettes were unique to their host[37]. This indicates that cassette biogenesis is a constant and efficient process in these species. Furthermore, even if many of the genes present in the cassettes have no counterparts in the databases, those that have homologues of bacterial origin show no bias for a specific group[37,57], which suggests that both Gram-positive and Gram-negative species are sources for the recruitment of gene cassettes.

The acquisition and subsequent activity of integron gene cassettes offers a rapid method for bacterial innovation. The size of superintegrons and their ancient and dynamic nature is a reminder that the gene cassettes currently associated with superintegrons represent only a fraction of those that might have participated in the evolution of their host. Therefore, comparing the organization of superintegrons from recent and earlier isolates of a particular strain, and between recent isolates from different geographical locations and ecological niches, could yield valuable biological information. For example, there are over 200 O serotypes of *V. cholerae* that constitute isolates of ecological, geographical and temporal diversity. Clark *et al.* examined the global organization of superintegrons from 65 *V. cholerae* O serotypes by PCR and Southern blot analysis[61]. Extensive restriction polymorphisms were observed even among closely related isolates, suggesting appreciable genetic plasticity in the structure of superintegrons and their microevolution through integrase-mediated gene acquisition and loss, and through cassette rearrangements[61]. A recent microarray-based comparative genomic analysis of different *V. cholerae* isolates showed that the O1 and O139 isolates contained more than 95% of the integron cassettes found in the superintegron of the O1 serotype N16961, whereas non-O1, non-O139 isolates contained only 50–75% of these cassettes[62]. The superintegron organization of two *V. cholerae* strains suggests that cassettes can be mobilized in clusters; the cassettes in positions 1 to 4 of the superintegron of strain 569B are found in the same order in the superintegron of strain N16961, but the cluster occupies a different relative position (positions 79 to 82 — a displacement of more than 40 kb). However, it is not known whether these differences represent a true group-mobilization event or temporal differences in cassette acquisition. In addition, there is evidence that not all of the *attC* sites associated with superintegron gene cassettes are equally functional; some sites contain mutations or deletions that could render them non-functional. As such, the movement of these gene cassettes would have to be coordinated with the movement of gene cassettes possessing functional *attC* sites. Collis and Hall demonstrated that integron gene cassettes are excised as covalently closed circles, and they observed differences in the resulting recombination products[63]. Some cassettes could be mobilized as individual units, and others could be excised in tandem with another cassette. Whether such cassette 'hitch-hiking' is by design to ensure the simultaneous transmission of genes is unknown.

The gene-cassette content of superintegrons also appears to vary extensively among isolates of other *Vibrio* species. Analysis of the superintegrons from two isolates of *V. vulnificus* indicates a similar number of gene cassettes: 188 in strain YJ016 and 211 in strain CMPC6 (REF. 33). Interestingly, although the two integrase genes differ by only a single amino-acid substitution, only 20% of the cassettes are shared, and only six of these are identical[33]. Furthermore, our analysis of strain CMPC6 revealed that the gene cassettes closest to the integrase gene (with a single exception) were not found in YJ016. This arrangement suggests that these cassettes might be derived from more recent acquisition events and, as such, are unique to the different isolates.

Integron evolution

Mobile integrons and superintegrons have identical structural organizations, and the antiquity of superintegrons suggests that they could be ancestral to mobile integrons. This relationship led to the proposal that mobile integrons evolved from superintegrons through the entrapment of *intI* genes and their cognate *attI* sites by mobile elements, such as easily and randomly assembled compound transposons. It was recently observed that 12 different resistance cassettes carry an *attC* site almost identical to the *attC* sites located in the superintegrons of *Xanthomonas* and *Vibrio* species[38]. It is tempting to speculate that once mobile, the subsequent harvesting of cassettes from various superintegron sources led

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Minimum inhibitory concentration
The lowest concentration of an antibiotic that inhibits growth of the organism.

to the 'creation' of contemporary mobile integrons, as reflected by the diversity of *attC* sites associated with the gene cassettes of mobile integrons.

The proposal that mobile integrons evolved from superintegrons has recently received support from two observations. The first is the discovery of a plasmid associated with trimethoprim resistance in *V. salmonicida*. Sørum and collaborators observed that trimethoprim resistance was due to a *dfrA1* gene cassette associated with a novel integron class (according to its *intI* gene) (H. Sørum *et al.*, unpublished data). The *dfrA1* gene cassette, commonly found in both class 1 and class 2 mobile integrons, is located at the second position of eight gene cassettes in the *V. salmonicida* superintegron. This new integron class has two features that are consistent with our evolutionary model. First, the integrase gene clearly originated from a *Vibrio* species or a closely related genus — the gene is 74% similar to the genes encoding the integron-associated integrases in *V. cholerae* and *Vibrio mimicus*. Second, the *attC* sites of three out of the seven other cassettes are structurally similar, a characteristic which, until now, has been associated only with superintegron gene-cassette arrays. As this integron is in close proximity to insertion sequences, it is tempting to speculate that this element corresponds to an intermediate structure in the evolution of the mobile integrons found in the genomes of clinical isolates. The creation of a mobile integron that imparts a multiple-antibiotic-resistant phenotype to its host organism could be achieved by the acquisition of additional resistance cassettes, taking advantage of the mobility of the assembled compound transposon and multiple transfer events.

The second piece of supporting evidence comes from our demonstration that a mobile integron can directly recruit the gene cassettes of a superintegron and acquire an antibiotic-resistant phenotype[59]. The recruitment of superintegron gene cassettes was shown to be a random process. By applying a selective pressure for the development of antibiotic resistance, a novel chloramphenicol acetyltransferase gene cassette (*catB9*) was discovered in the superintegron of *V. cholerae*. Taking into account previous results demonstrating that superintegron gene cassettes are substrates for integrases from class 1 integrons when present on a high-copy-number plasmid[30], these results demonstrate that environmental conditions, including the presence of antibiotics, dictate which of the randomly recruited cassettes are retained in the mobile integrons of clinical isolates.

Two additional resistance cassettes — CARB-7 and CARB-9, which encode novel carbenicillinases — have been identified in the superintegrons of two environmental isolates of *V. cholerae*[64,65]. Interestingly, they confer different levels of β-lactam resistance on their host, with the minimum inhibitory concentration measured for an isolate containing CARB-9 being significantly less than that for an isolate containing CARB-7, depending on the β-lactam used[65]. These novel carbenicillinases belong to a subgroup of the class A β-lactamase family, the RSG carbenicillinases[66], which is characterized by an RSG amino-acid triad in positions 234 to 236. This subgroup of carbenicillinases is encoded by gene cassettes and is associated with either a VCR-like *attC* site of 123 bp or an *attC* site of 103 bp. With the exception of CARB-7 and CARB-9, RSG carbenicillinase cassettes have been identified in class 1 integrons from a variety of bacterial hosts (*Pseudomonas aeruginosa*, *Salmonella enterica* serovar Typhimurium, *Proteus mirabilis* and *V. cholerae*). Noticeably, phylogenetic analysis revealed that the oldest cassettes were those associated with a VCR-like *attC* site[65]. Therefore, it is possible that an ancestral RSG carbenicillinase originated in *V. cholerae* or a closely related species. If correct, this would support our hypothesis that gene cassettes found in contemporary mobile integrons were recruited from various superintegron sources.

It is also likely that the assembly of complex mobile integrons (mobile integrons carrying more than two resistance cassettes) occurred by recombination between different mobile integrons, rather than as a result of successive recruitment events from the superintegron gene-cassette arrays in different environmental species. There are niches that could favour a high level of exchange of cassettes between different mobile integrons. The characterization of a remarkable collection of gene cassettes, integrons and plasmids circulating in a single wastewater treatment plant[67], and the recent demonstration that aerobic Gram-positive corynebacteria can act as a reservoir of class 1 mobile integrons in poultry litter[28], indicates the existence of environments, other than the obvious clinical settings, in which the exchange of gene cassettes between mobile integrons could be a common process.

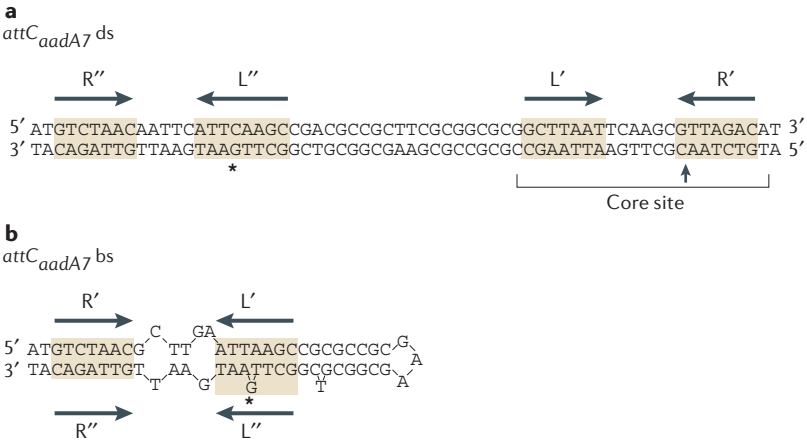

Figure 3 | **Integron recombination sites. a** | The sequence of the double-stranded *attC* site in the *aadA7*-encoding cassette ($attC_{aadA7}$ ds). **b** | The proposed secondary structure for the $attC_{aadA7}$ bottom strand (bs). The inverted repeats L′ and L″, and R′ and R″ are indicated with arrows. The asterisk shows the position of the protruding G base present in L″ relative to L′. The putative integrase (IntI1)-binding domains, as defined by Stokes *et al.*[81], are marked with yellow boxes. The vertical arrow indicates the crossover position. The secondary structure was determined using the MFOLD[88] online interface, at the Pasteur Institute.

Recombination reactions in integrons

As defined earlier, integrons contain a gene encoding an integrase belonging to the integrase family of tyrosine recombinases (for a review see REF. 68), as well as

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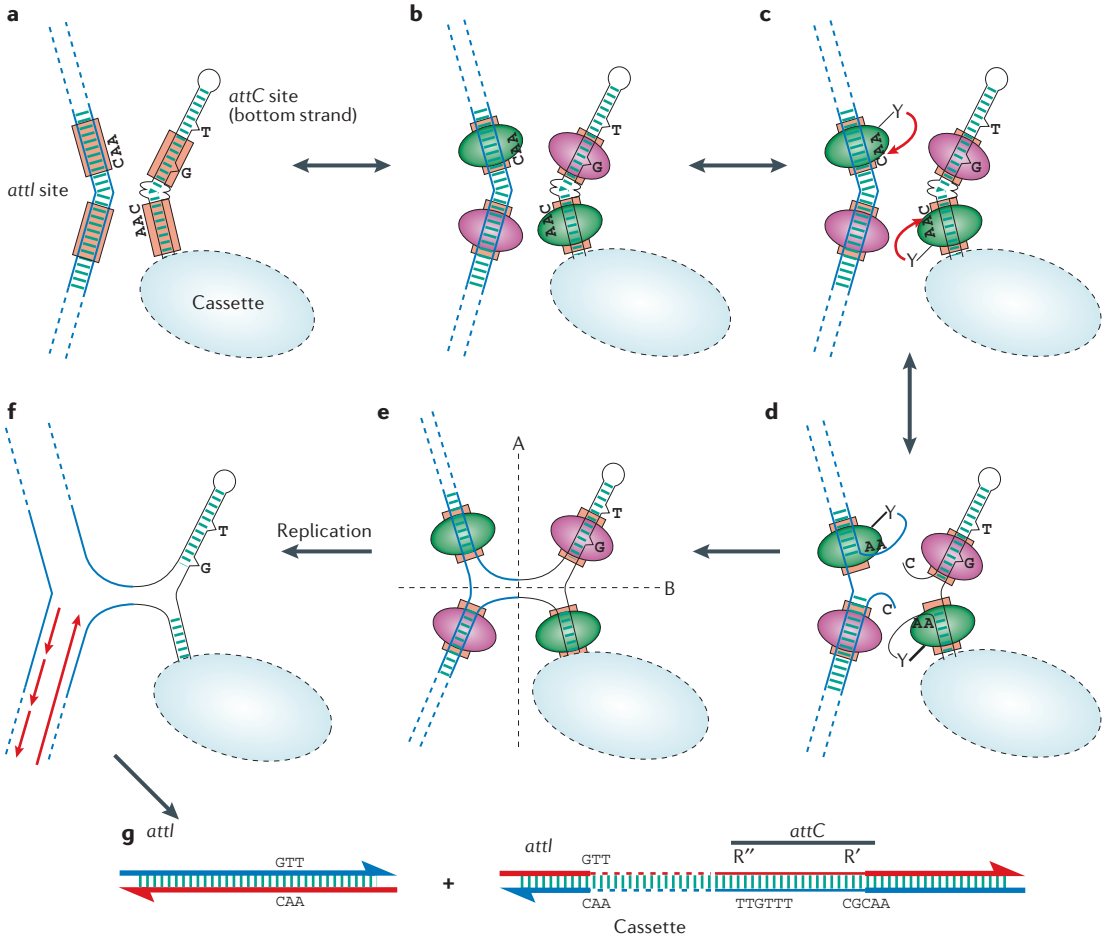

Figure 4 | **A model of integron-mediated recombination.** This model uses a single-stranded (ss) *attC* substrate folded through pairing of the imperfect palindromic sequences. The steps are identical to classical site-specific recombination steps catalysed by other tyrosine recombinases, up to the Holliday-junction intermediate. **a** | The R and L IntI binding domains of the *attI* double-strand and *attC* bottom-strand (bs) core sites are boxed. For simplicity, only the cassette bs is depicted.) **b** | Binding of the four IntI monomers to the core sites. The green monomers correspond to the attacking subunits and the pink correspond to the non-attacking subunits (see FIG. 5). **c** | The attacking subunits cleave the DNA strand with catalytic tyrosine 302 to form 3′-phosphotyrosine linkages (**d**) and release free 5′-hydroxyl groups. **e** | The 5′-hydroxyl groups undergo intermolecular attack of the partner phosphotyrosine to complete the exchange of one pair of DNA strands between the two substrates and form a Holliday-junction intermediate. Classical resolution through the A axis would reverse the recombination to the original substrates, whereas resolution through the B axis, giving rise to covalently closed linear molecules, would be abortive. The non-abortive productive resolution necessitates a replication step shown in (**f**,**g**) . The red lines in the last panel show the newly synthesized strands.

an associated primary recombination target called the *attI* site. Integron-encoded integrases mediate recombination involving two sites — the specific *attI* site and the gene-cassette-associated *attC* site — and can recombine distantly related DNA sequences. This last point is reflected by the sequence differences observed between the *attC* sites from different gene cassettes.

The recombination reactions of integrases from mobile integrons (classes 1, 2 and 3) and several superintegrons have been studied to different extents. The paradigm for integron recombination is undoubtedly the class 1 integrase, IntI1. Indeed, since its initial characterization in 1990 (REF. 69), IntI1-catalysed recombination has been studied extensively by many groups. The ability of IntI1 to recombine distantly related *attC* sites has been demonstrated in several studies, and five recombination reactions have been established. Three correspond to the different recombination events possible between the normal recombination sites (*attI* × *attC*, *attC* × *attC* and *attI* × *attI*), and the other two correspond to inadvertent recombination between either an *attI* or *attC* site and non-specific GTT-containing sequences[70–75]. Studies have shown that the recombination events involving two *attI* sites are significantly less efficient than the reactions involving two *attC* sites, and recombination between *attI* and *attC* is the most efficient reaction[22]. Although integron-encoded integrases can recombine structurally different *attC* sites, recombination at the *attI* site is specific. However, IntI1 and the integrase from *N. europaea* (IntINeu) have been shown to be able to delete cassettes in the first position of heterologous *attI* sites, albeit at low frequencies[21,41].

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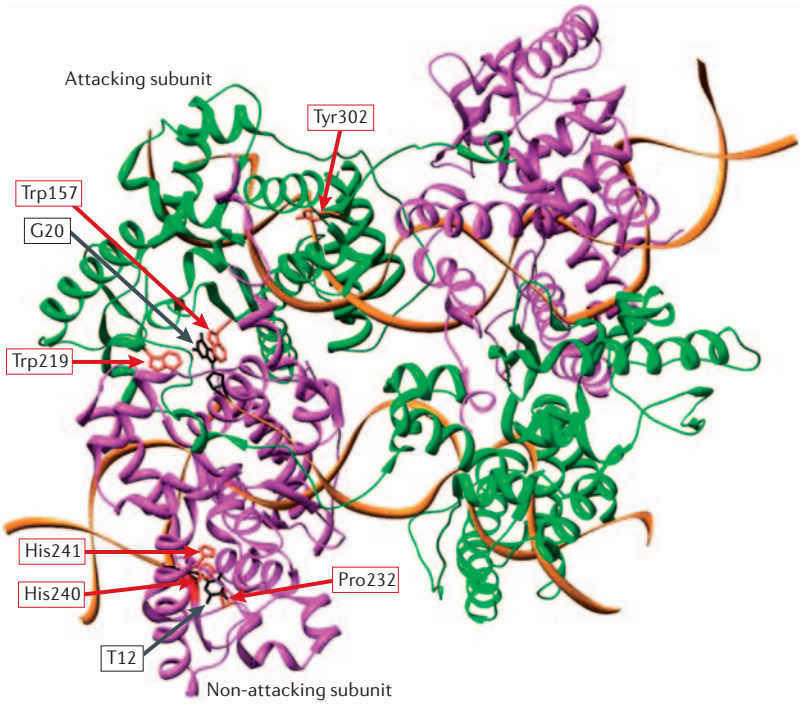

Figure 5 | **Architecture of an integrase–*Vibrio cholerae* repeat (VCR) synapse.** Four *Vibrio cholerae* integrase (VchIntIA) molecules are bound to two antiparallel VCR repeat-bottom-strand (VCRbs) helices to form the active synapse[86]. The attacking subunits are shown in green, and the non-attacking subunits are shown in pink. The extrahelical base T12 (black) is stabilized by *cis* interactions and is involved with DNA site recognition. A second extrahelical base, G20 (black), is buried in a deep hydrophobic pocket in subunits that are bound to the other VCRbs helix to form a set of *trans* interactions. The non-symmetrical protein–protein interfaces between VchIntIA molecules yield an overall synapse that has two-fold symmetry. The amino-acid residues interacting with the extrahelical T12 and G20 bases are indicated in red. The catalytic tyrosine residue in one of the attacking subunits is also indicated (Tyr302).

The cassette deletion and integration activities of the integrase from *S. oneidensis* (IntISon), and IntINeu, both of which are associated with a small number of gene cassettes or no cassettes at all, have been demonstrated but not quantified[40,41]. In addition, the integrative recombination reaction (*attI* × *attC*) catalysed by the superintegron integrase of *P. stutzeri* (IntIPstQ) has been demonstrated[76]. Deletion of gene cassettes owing to recombination between the *attC* site in the *aadB*-encoding cassette ($attC_{aadB}$) and the *V. cholerae* superintegron cassette *attC* site has also been established for VchIntIA[38].

As discussed above, superintegron *attC* sites are highly homogeneous and species-specific, whereas the *attC* sites in mobile integrons are highly variable in length and sequence. One hypothesis to explain the narrow range of heterogeneity observed for the *attC* sites of superintegron gene cassettes is that the superintegron integrase has narrower substrate specificity than the mobile integron integrase. However, a comparative study of the recombination activities of IntI1 and VchIntIA[77] has shown that although the diversity of *attC* sites recognized by VchIntIA is narrower than the diversity recognized by IntI1, VchIntIA-mediated recombination is not limited to *attC* sites related to the VCR-containing gene cassettes of *V. cholerae*. This indicates that the nearly exclusive presence of cassettes carrying a VCR *attC* site in the *V. cholerae* superintegron is not due to the restricted range of recombination-site recognition by IntIA[77]. Another puzzling observation was that the formation of co-integrates through VchIntIA-mediated recombination between the *V. cholerae attI* site and the VCRs occurred at a 2,000-fold higher rate in *V. cholerae* than in *Escherichia coli*, whereas the rate of IntI1-mediated *attI1* × VCR recombination was identical in both species[77]. These results were unexpected, as the VCR gene cassettes and *attI*Vch recombination sites are the natural substrates of VchIntIA. Taken together, these results indicate that substrate recognition and the recombination reactions catalysed by the integrase VchIntIA might differ from the paradigm for class 1 mobile integrons.

It was assumed that site-specific recombination by integron-encoded integrases did not require any accessory proteins. However, as an *in vitro* recombination assay has yet to be successfully developed, this assumption is open to debate. One possible explanation for the observed low level of integration in a reconstituted system in *E. coli* is that, at least for the integration process, VchIntIA requires an accessory protein that is either absent or too divergent to be fully functional in the reconstituted system. In the case of the IntI1 integrase, if such an accessory protein is required for integration at the *attI1* site, the evolutionary constraints exerted on mobile integron systems, which are carried on mobile elements and selected to be operational in multiple hosts, could explain why IntI1 is able to recombine at the same rates in *E. coli* and *V. cholerae*.

Evidence for a novel recombination pathway in integrons. Another important discovery relates to the nature of the recombination substrates. A unique trait of the integron recombination system is the structure of the recombination sites. The typical core recombination sites for tyrosine recombinases consist of a pair of highly conserved 9–13 bp inverted binding sites separated by a 6–8 bp central region[78]. The *attI* sites differ from this canonical organization in that one of the putative binding sites in the core site is always degenerate, and the central region differs greatly between different *attI* sites. However, *in vitro* experiments have shown binding of IntI1 to four regions of the double-stranded *attI1* site, two of which correspond to the core site. The other two regions form direct repeats located 5′ to the core site[79,80].

The structure of the *attC* site is more complex. It consists of two potential core sites — R″–L″ and L′–R′ (REF. 7) (also termed 1L–2L and 2R–1R, respectively[81]) — separated by a central region (FIG. 3a). Only L′–R′ is recombinogenic. If the central region is highly variable in sequence and size (20–104 bp), the site always has a palindromic structure, and all structurally different *attC* sites have the potential to form cruciform structures by the self-pairing of both strands[37,81,82] and are efficiently recombined by IntI1. In 1999, Francia and colleagues

Co-integrates
The structural fusion of replicons through a recombination reaction.

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Holliday junction
A point at which the strands of two double-stranded DNA molecules exchange partners, which occurs as an intermediate in genetic recombination.

demonstrated that purified IntI1 interacted specifically with the bottom single strand of $attC_{aadA1}$ DNA from the *aadA1* cassette but not with the double-stranded $attC_{aadA1}$ site[83]. Sundström and colleagues confirmed this important observation and identified several important elements in the $attC_{aadA1}$ sequence that function as recognition determinants for IntI1 and have important roles in the potential secondary structure of the *attC* site[84].

In most circularized gene cassettes, self-pairing on the same single strand can be structurally extended up to the R′ and R″ sequences, which usually have 9–11 consecutive complementary nucleotides[37,74]. This self-paired stem resembles an almost canonical core site consisting of the L″–L′ duplex, an unpaired central region followed by an R″–R′ duplex (FIG. 3b). Recognition and recombination of such a structure with a canonical double-stranded *attI* site by an integron integrase would lead to the formation of a Holliday-junction intermediate that could be resolved by a replication step.

Recently, this hypothetical model has received support from the *in vivo* demonstration that only one DNA strand of the *attC* site is recombined by IntI1. Indeed, using an *in vivo* recombination assay based on a conjugation system that independently delivers either the top strand or the bottom strand of the different integron recombination sites into a recipient strain, it was found that the recombination rates of the different substrates varied across a wide range[85]. The rate of recombination with DNA encoding the double-stranded *attI* site (carried on a plasmid), following conjugative transfer of DNA encoding the bottom strand of the *attC* site, was determined to be 1,000-fold higher than the rate observed following conjugation of the top strand of the *attC* site. Furthermore, the recombination rate after delivery of DNA encoding the bottom strand of the *attC* site was identical to that observed in a classical assay using double-stranded *attC* and *attI* sites present on plasmids and co-maintained in bacterial cells expressing the IntI1 protein. Conversely, only the *attC* sites seemed to recombine in single-stranded form, as recombination following conjugative transfer of either the top or bottom strands of the *attI* site was observed to occur at a rate 1,000-fold lower than that measured in the same classical assay.

Together, these results support a recombination model for the insertion of integron gene cassettes at the double-stranded *attI* site that involves only the *attC* bottom strand, folded in a stem–loop structure, and resolution of the generated Holliday-junction intermediate through replication (FIG. 4). The recent solution of the crystal structure of an integron integrase tetramer bound to DNA encoding the bottom strand of the *attC* site provides additional support for this model[86] (FIG. 5). The structure shows that only two out of the four subunits have an active conformation, and the other two subunits are involved in specific interactions with two nucleotide bases that protrude from the *attC* bottom strand self-paired structure. Remarkably, these two extrahelical bases are conserved among all the *attC* sites characterized so far[37,81,85] and are thought to have a major role in recognition of the *attC* bottom strand[84]. The structure also shows that the IntI-specific additional segment is directly involved in the recognition of these two extrahelical bases.

The selective advantages that have led to the development of these single-stranded DNA recombination sites and processes remain elusive. It is possible that this phenomenon could be linked to gene dissemination by horizontal transfer, which, in many cases, has a single-stranded stage in the DNA transfer process (as has been demonstrated in the processes of conjugation, and natural transformation in bacteria and in filamentous phage[87]).

Conclusions

The recruitment of exogenous genes is the most efficient means by which bacterial species can survive various environmental challenges, including exposure to antimicrobial compounds. Integrons provide bacteria with a gene-capture system perfectly adapted for the challenges of multiple-antibiotic treatment regimes. However, with the discovery of superintegrons, and the thousands of gene cassettes associated with integrons in the genomes of environmental bacterial species, the importance of these elements clearly extends beyond the phenomenon of antibiotic resistance. The recruitment of integron gene cassettes endows the recipient bacteria with new proteins and new enzymatic functions, potentially giving the organism an adaptive evolutionary advantage. Both experimental and phylogenetic data suggest that superintegrons could be the ancestor of both mobile integrons and the resistance-gene cassettes observed in the genomes of bacterial isolates of clinical significance. Nevertheless, many pertinent questions about this system remain unanswered. In particular, there is a need to elucidate the mechanisms underpinning the recombination process, the formation of new gene cassettes and the dynamics of gene-cassette exchange in complex bacterial populations.

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Acknowledgements
Work in the Mazel laboratory was supported by the Institut Pasteur, the Centre National de la Recherche Scientifique (CNRS), the Programme de Recherche en Microbiologie (from the Ministère de l'Education Nationale, de l'Enseignement Supérieur et de la Recherche; MENESR), the Institut Français de Recherche pour l'Exploitation de la Mer (IFREMER) and the European Union Collective Research on Aquaculture Biofouling (CRAB) consortium and Network of Excellence EuroPathoGenomics).

Competing interests statement
The author declares no competing financial interests.

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